Treetable / Living Cloud — Technical One-Pager

Evidence layer for ecological sensing: every field reading carries a witnessed, replayable receipt. Non-invasive sensing only — no attachment, no probe insertion, consent-gated actuation.

LIVE PRODUCT: replay-verified compute API
connectome-compute-api.terexmaps.workers.dev
- Deterministic LIF simulation over 15 real EM-derived connectomes (C. elegans, Drosophila, hemibrain, medulla ~40k neurons, mouse retina, macaque, human segments and more)
- Every task returns task_said + trace_digest; POST /verify replays the task and confirms byte-identical digests. Compute you can check — every task ships its own proof.
- TypeScript port verified bit-identical to the Python reference engine (930/930 checks, Q24 fixed-point, banker's rounding, int64-LE digests, BLAKE2s content addressing)

EDGE / PROTOCOL LAYER
- Receipt chain: content-addressed evidence bundles, quorum witnessing, replay bundles
- EcoCell One node: XIAO ESP32-S3 + E22-400M22S Reticulum radio, BME280/BH1750/AD7746, TP4056 charge + MT3608 boost, SMA antenna, consent-gated emit path — manufacturable Gerber/drill/BOM/pick-and-place package complete (Seeed-ready)
- BYOD mode: the same evidence path runs on phones/Raspberry Pis — no custom hardware needed

MEASURED POSITIONING
- Sparse temporal computation benchmarked against dense O(n^2) baselines; honest result: advantage derives from sparsity — biological topology provides identity, provenance and replayable structure. Refused-lane results are published, not hidden.
- License: AGPL-3.0-or-later. Repository verified end-to-end (90/90 e2e checks).

TRACTION
- API live in production; verification-audit offers out to Cortical Labs + FinalSpark
- NSF SBIR/STTR Project Pitch #00131003 submitted; Crowd Supply under review; Elemental Impact application submitted